TADPOLE Closed Loop Control

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June 15, 2025

1 Introduction

Creating a fast, responsive, and robust controller for TADPOLE is essential to our success both with TOAD and with the CLPC Throttleable Hotfire challenge. The challenge rules state the following: "Hotfire Primary Propulsion Device for at least 10 seconds of controlled thrust (not including a transient start-up period). [...] demonstrate controlled & constant nominal thrust above 500 lbf for 4 seconds, smoothly throttle down to below 40% of that nominal thrust and hold constant for at least 2 seconds, then return to the same nominal thrust and hold constant for at least 1 second." The error margins allowed are the following:

- ± 5% of nominal thrust at every point in the curve
- ± 0.5s window of leniency during throttle up and throttle down periods
- Thrust must NEVER surpass 40% of nominal thrust during the low thrust hold.

These error margins are tight, and open loop valve control, although it might work on its own, proves to not be the most reliable option. As such, building a feedback loop controller gives the team the best chance of success in this challenge, and in the future, when TADPOLE will receive fast-changing commands as it tries to vertically land TOAD.

2 Thrust Curve

For a successful completion of the CPLC milestone, the team created the following thrust curve that adhered to all milestone rules, and incorporated smooth edges to give the controller the best chance of success. The final thrust curve used during the throttle attempts is the following [Figure 1]: This thrust curve pushed the limits of our designed operating range, by targeting a maximum thrust of 590 lbf, and a minimum thrust of 205 lbf.

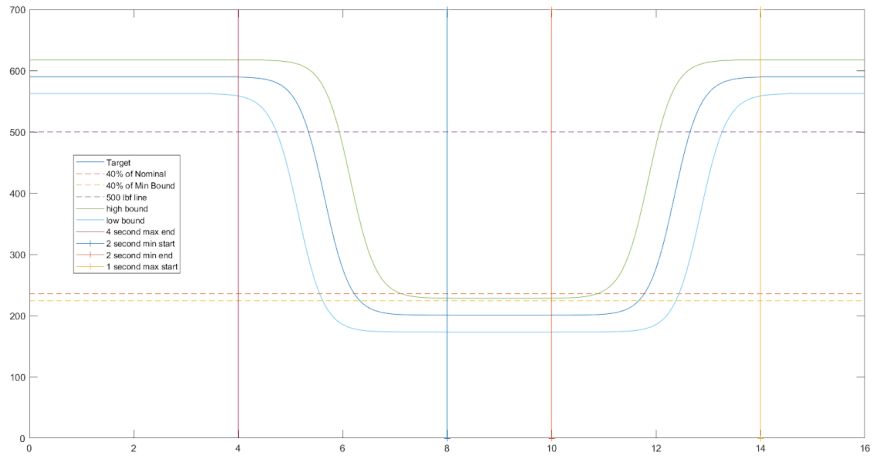

Figure 1: Target Thrust Curve

3 Plant Model

We have developed a plant model of TADPOLE, by expanding on the work presented by Y. C. Lee and Summerfield in their papers on chemical rocket engine stability and control. Their work was also re-derived from first principles and expanded to include actuators to control mass flow into the chamber, instead of mass flow control by using orifices. The differential equations that govern the model are the following:

$$\frac{d\dot{m}_{ox}}{dt} = (P_{ox} - P_c - (\Delta P_{i,ox} + \Delta P_{l,ox}) \cdot \dot{m}_{ox}{}^2) \cdot \frac{gA}{l_{ox}}$$
$$\frac{d\dot{m}_{ipa}}{dt} = (P_{ipa} - P_c - (\Delta P_{i,ipa} + \Delta P_{l,ipa}) \cdot \dot{m}_{ipa}{}^2) \cdot \frac{gA}{l_{ipa}}$$
$$\frac{dP_c}{dt} = \frac{RT_c}{V_c} \cdot (\dot{m}_{ox} + \dot{m}_{ipa} - \frac{gA_t}{c^*} P_c)$$
$$\frac{dP_{ox}}{dt} = \frac{RT_c}{V_c} \cdot (231/60 \cdot C_{v_{ox}} \sqrt{\rho_{ox} \rho_{H_2O} (P_{tank} - P_{ox})} - \dot{m}_{ox})$$
$$\frac{dP_{ipa}}{dt} = \frac{RT_c}{V_c} \cdot (231/60 \cdot C_{v_{ipa}} \sqrt{\rho_{ipa} \rho_{H_2O} (P_{tank} - P_{ipa})} - \dot{m}_{ipa})$$

where:

$$\Delta P_i \rightarrow \frac{1}{2\rho_f g(C_d A_i)^2}$$
$$\Delta P_l \rightarrow C_{fric} \cdot \frac{l}{2dg\rho_f A^2}$$

represent the pressure drops due to the injector and line friction. P_{ox} and P_{ipa} represent the pressures downstream of the valves. A is the line cross-sectional area. Figure 2 shows a simulated

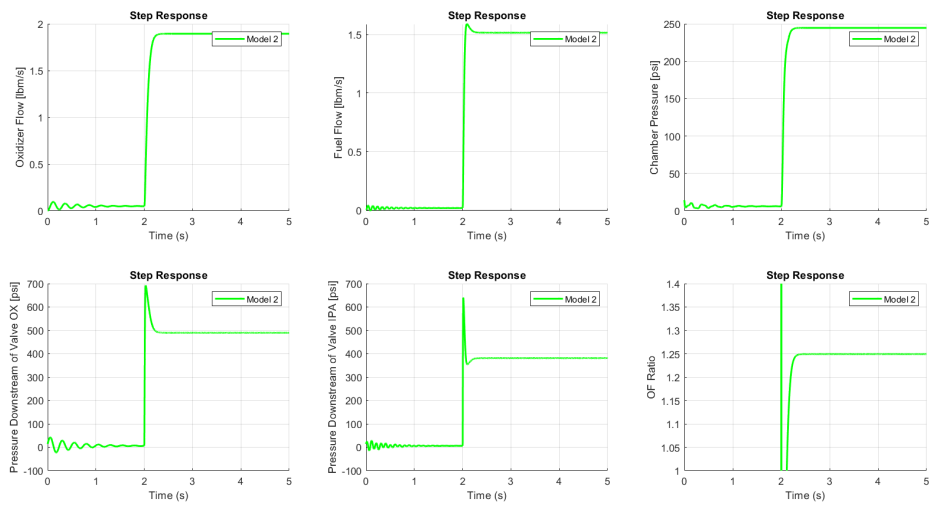

Figure 2: Step Responses for Mass Flows and Chamber Pressure.

startup at T+2 seconds at 250 psi chamber pressure, using tank set pressures of 700 psi (similar to those used during the testing campaign). Note that the model fails to properly reproduce the water hammer effect seen during actual startups, as well as failing to reproduce some oscillations due to the droopy tank pressure regulators used during the testing campaign. Figure 3 shows our Simulink implementation of the Plant Model. Initially, actuator position error was assumed negligible, and we only took into account dead-time delays and valve response times. As will be explained later, valve tracking errors and stickiness turned out to be a significant issue during the campaign. The model shown here includes the last minute addition of dynamic actuator models, using a simple set of differential equations and a minimum torque to move the valve to model the sticking. The controller provided by oDrive for the motors was also recreated in Simulink in order to better understand the valve's behavior. The equations for the dynamic valve model are the following:

$$\frac{d\theta}{dt} = \omega$$
$$\frac{d\omega}{dt} = \frac{\tau}{I} - k\omega$$

in which τ is reduced to zero if below an experimentally determined threshold, to reflect the torque needed to overcome the valve's static friction. $k\omega$ is a dampening parameter chosen to match the valve's real world behavior. Overall, this model is extremely simplified, but proves enough to replicate the valve's behavior.

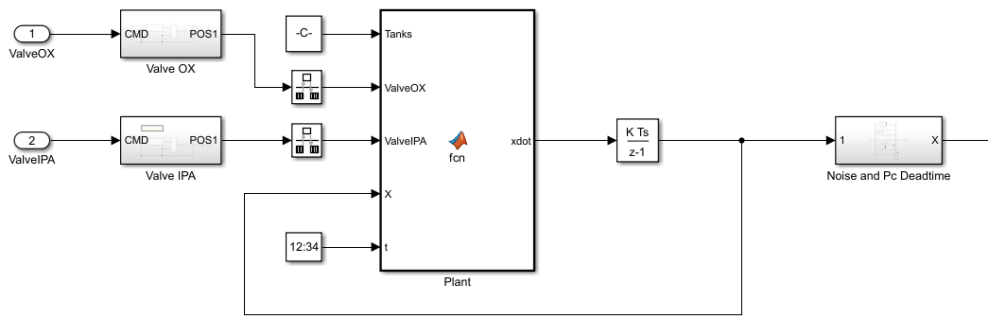

Figure 3: Simulink Implementation of Plant

4 Controller Design

Our goal is to get a fast, responsive and robust controller for TADPOLE, that is capable of tracking a reference thrust and a constant mass flow ratio of 1.2. To accomplish this, the team had previously developed an Open-Loop Controller. Open-Loop controllers are great mathematical models and a great baseline for us. However, they fail to account for important factors such as potential model errors, dynamics, and delays. For this, we use a feedback loop to make real-time corrections. Our proposed controller design is not ordinary and can be best described as a feed-forward controller with feedback trim. Figure 4 shows a block diagram of our controller implementation.

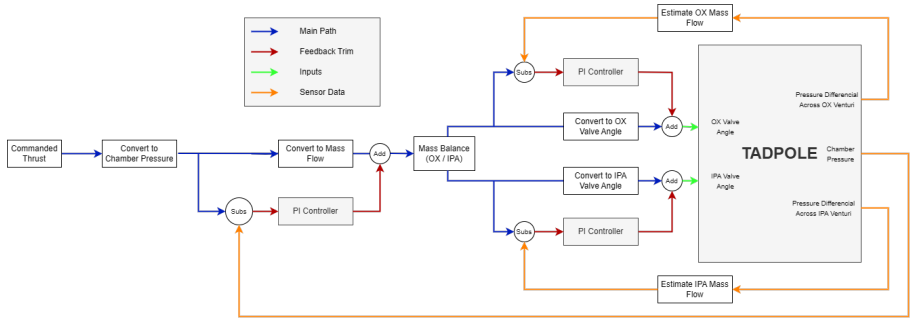

Figure 4: Block Diagram Implementation of Controllers

The structure includes a Chamber Pressure Feedback controller, cascaded by 2 parallel Mass Flow controllers, which are in charge of keeping our mass flow ratio reasonably constant. Our feed-forward terms carry what should be the correct valve angles, and we use our feedback controllers to integrate the error signals and trim these angles down to correct values. This control structure allows us to efficiently control our engine, and it is a very safe way to start, since most of the control action comes from the feed-forward. Figure 5 shows an example of this division of control authority during a cold flow test of TADPOLE. This reliance on our model allows the feedback action to serve only as a trim, correcting for unaccounted errors, instead of having to build a control signal from the ground up. The PI loops are tuned such that the mass-flow controllers are "fast loops" and the Chamber pressure controller is considered a "slow loop".

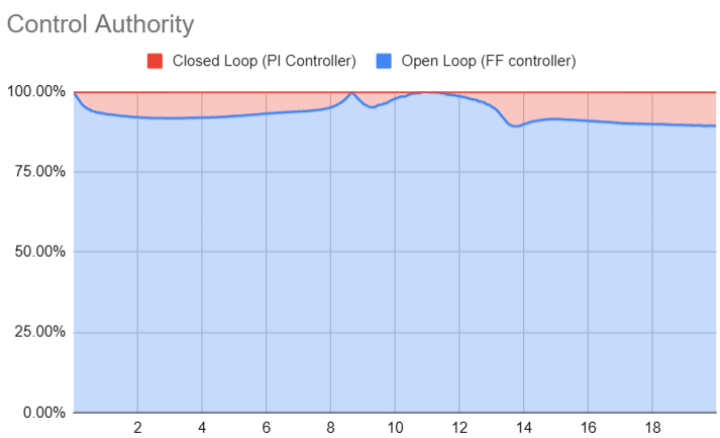

Figure 5: Control Authority Division during Cold Flow Test

The feedforward controller is implemented as follows: The controller receives a reference thrust signal T, which it then converts to a target chamber pressure P_c by using

$$P_c = \frac{T}{A_t c_f}$$

This chamber pressure then becomes a target total mass flow $\dot{m}$ by

$$\dot{m} = \frac{P_c A_t}{c^*} g$$

Note the factor of g due to the conversion from pounds force to pounds mass. This total mass flow gets divided into the required oxidizer mass flow and fuel mass flow by using our target OF ratio of 1.2:

$$\dot{m}_{ox} = \frac{\dot{m}}{1 + OF} \cdot OF$$
$$\dot{m}_{ipa} = \frac{\dot{m}}{1 + OF}$$

For the final step of turning target mass flow into a valve angle command, we derive a feedforward controller from our steady state plant dynamics (setting all the $\frac{d}{dt}$ terms equal to zero), and exploiting the conservation of mass relationships that show up. By combining this with the valve equation

$$C_v = Q\sqrt{\frac{SG}{P_{in} - P_{out}}}$$

and converting $Q = \dot{m}$ and $SG = \frac{\rho_{fluid}}{\rho_{water}}$, we end up with the following equation, used on both the oxidizer circuit as well as the fuel circuit. Note that for LOX, the fluid density is obtained real time by using sensor data and a look-up table relating temperature to density:

$$C_v = \frac{60}{231} \cdot \dot{m}_{trg} \cdot \sqrt{\frac{1}{\rho_{fluid}\rho_{H_2O}(P_{tank} - P_{back} - \Delta P_i \cdot \Delta P_l \cdot \dot{m}_{trg}{}^2)}}$$

where:

$$P_{back} \rightarrow \text{ Back pressure on the system.}$$
$$\Delta P_i \rightarrow \frac{1}{2\rho_f g (C_d A_i)^2}$$
$$\Delta P_l \rightarrow C_{fric} \cdot \frac{l}{2dg\rho_f A^2}$$
$$\dot{m}_{trg} \rightarrow \text{ Target mass flow rate}$$
$$C_v = \frac{\alpha}{1 + e^{\frac{\theta - \beta}{\gamma}}} \text{ where } \alpha, \beta, \gamma \text{ are constants identified during water flows}$$

Using this equation and the provided mapping function C_v to angle θ, we are able to find a target valve angle for a given thrust target. The C_v mapping function might seem a bit arbitrary (and in fact it is), but testing showed that a sigmoid function was a good curve fit for our valve angle to C_v mapping. It also provided good benefits (continuity, easily invertible, monotone increasing) that were not ensured by look-up tables or polynomial curve fits.

5 Waterflow Testing Campaign

To make sure our controller and hardware performed as intended, the team completed an exhaustive water flow testing campaign to confirm functionality of all aspects of the system, from PT readings, Throttle Box commands, valve tracking, controller functionality, among others. This campaign proved to be a challenge, and the team encountered difficulties in every aspect listed above. This section will focus on the problems encountered with the controller and the valves.
One of the central aspects of the feedforward controller is the ability to accurately match valve angles to flow coefficients. For this purpose, the manufacturer had provided a look-up table for us to use. Nevertheless, during the first round of waterflow tests, significant mismatches between manufacturer provided data and measured values were found. Figure 7 portrays the theoretical Angle to flow coefficient mappings for both a V30 valve and a V60 (the one the team supposedly had), as well as the experimental results.

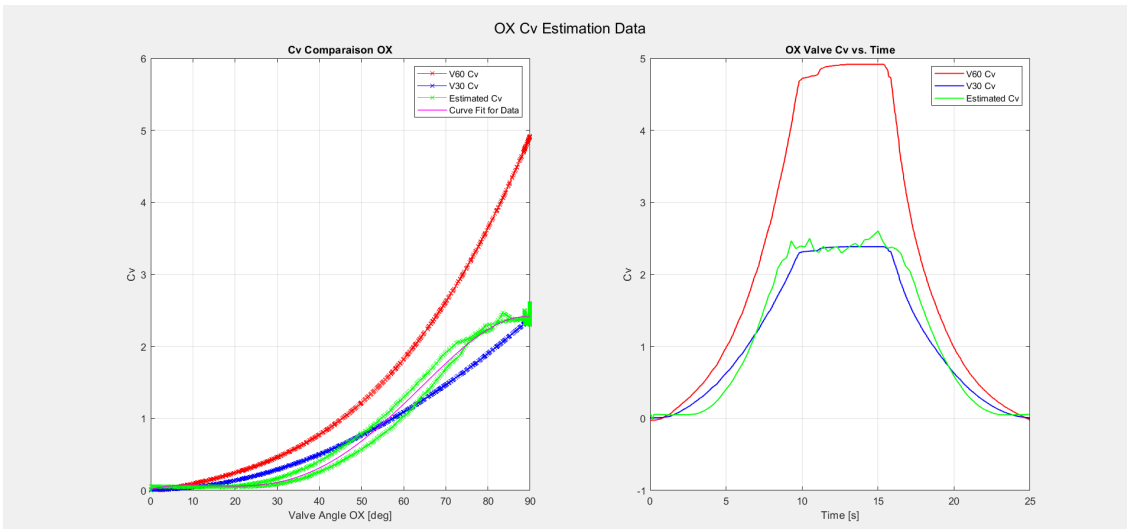

Figure 6: Flow Coefficient Estimation

It is interesting to note that our experimental results seem to align more with a standard ball port valve than a V-port valve. Figure 8 shows the flow patterns for 3 different valve port types. These issues were persistent and identical on both valves. Our math was rechecked and dimensional analysis was performed on every equation to make sure all the units checked out. The team then decided to move forward and create our own mapping for the valve's flow coefficients by performing angle sweep tests during a waterflow and measuring mass flow with a subcritical venturi. These tests produced the following flow coefficient curves [Figure 8].
These mapping functions held up during cold-flow testing with more challenging fluids, such as Liquid Oxygen, known for being highly compressible and for evaporating easily. Another issue encountered during waterflows was the feed-forward controller performance. Originally, the team attempted to use live downstream and upstream pressure measurements in the feedforward controller presented above, however, for reasons still unclear, this caused the controller to reach a sort of incorrect steady state, most likely due to the strong coupling between mass flows, and pressures

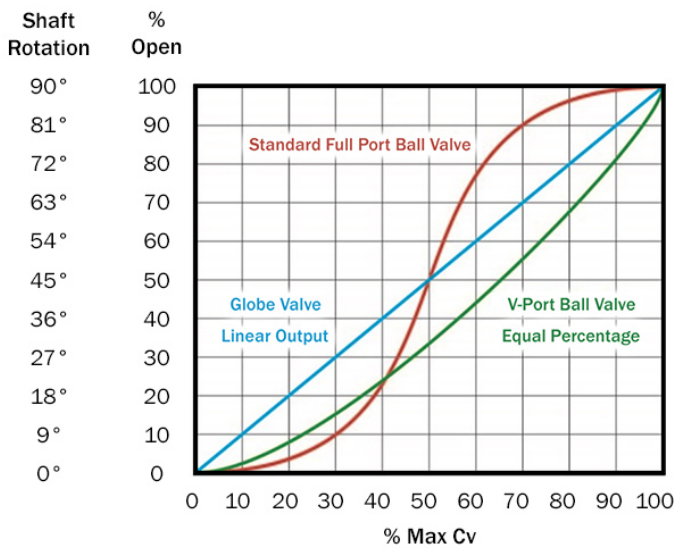

Figure 7: Flow patterns for different valve types

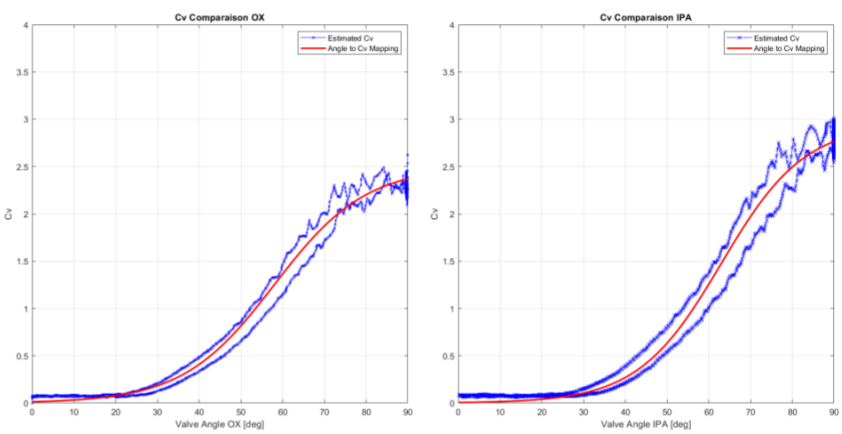

Figure 8: Flow Coefficient Mapping

across the valve. As such, the controller was switched to use predicted downstream pressures from the steady state plant dynamics. This also required a reformulation of the model dynamics, as we had to treat the section of pipe downstream of our valve setup during waterflows as a chamber with an orifice, so we could attempt to predict the pressure downstream of the valve and the pressure drop across of it. Finally, after successful verification of mass-flow tracking capabilities [Figure 9] the team moved forward with a Hardware In The Loop test, which attempted to test the Chamber Pressure control by mapping the measured mass-flows into operating values and feeding these

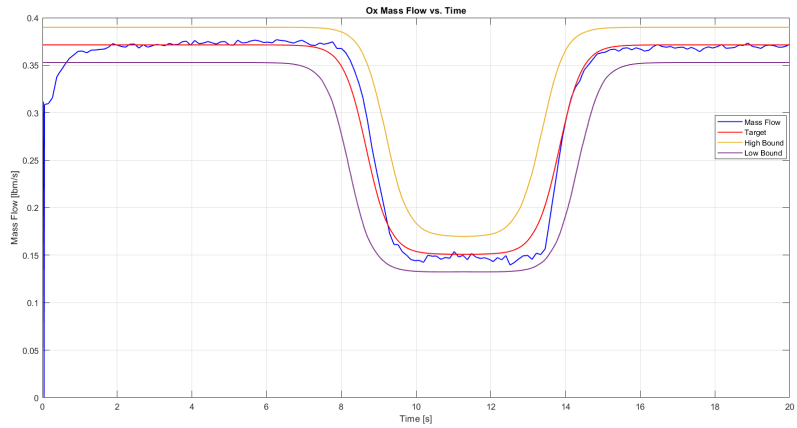

Figure 9: Waterflow Verification Test on OX Valve

values into the model, which was running live on a computer and feeding back simulated chamber pressure values. The team then incorporated a potentiometer to simulate pressure disturbances and test controller reaction to said disturbances. This test proved somewhat catastrophic however. Due to unresponsive valves, the team encountered coupling between the mass flow and the chamber pressure controller, that produced oscillatory behavior in the valves and integral windup which was not covered by our integral clamping methods. An example of such behavior during the HITL test is presented in Figure 10, along with commanded valve angle and measured valve angle.

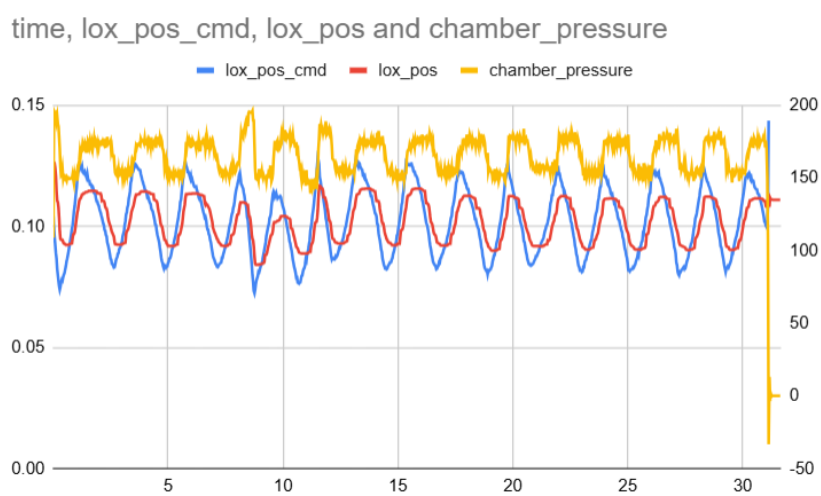

Figure 10: Oscillations on HITL Test

This behavior is what ultimately prompted the last minute addition of dynamic actuator mod-

eling. This also caused the team to run the entirety of the Hotfire campaign without activating the Chamber Pressure trim controller, to avoid potential issues which could prove dangerous on a firing test article.
Overall, the waterflow testing campaign proved really useful. An immense quantity of bugs and mistakes were found and fixed on time for cold flows to begin.

6 Propellant Coldflow Campaign

The cold-flow campaign was an important step, especially from a hardware perspective, as it validated the integration of both the DAQ system, the fully assembled throttle box with all measurements active, and valve installation on the test stand. Nevertheless, it also proved challenging from a controls perspective. The first step for the team was cross-checking the flow coefficient mapping previously created during water flow testing. Fortunately, this checked out for both propellants, and required no changes. Another important step was the creation of a new auto-sequence for hot-fire startup. Previously, the static fire campaign was conducted with two cavitating venturis located downstream of the run valves, which allowed for de-coupling of the downstream behavior and a simplified start-up sequence. However this time around, the throttle valves replaced the cavitating venturis, removing the decoupling and complicating the sequence. For start-up, the team decided to proceed by manually selecting starting angles for the throttle valves to be set at, The run valves were then opened, and at about T+1 second, when startup transients passed, we had a "controller handoff", where the closed loop controller took over and brought the valves into their optimal positions.
Selection of starting valve angles proved surprisingly challenging, as valve hysteresis was a persistent problem throughout the testing campaign (This effect is visible in Figure 8, identified by two distinct lines in the scatter plot, one represented flow coefficient on opening and one on closing). To accomplish this task to the best of our abilities, we used both our feedforward predictions and plots from flow coefficient angle sweeps to choose the best one. Figure 11 is an example of a Valve Angle vs. Mass Flow plot generated for this purpose, with the target mass flow represented with a red line.

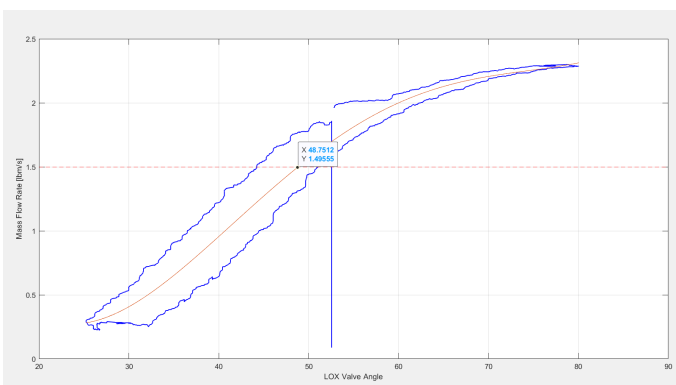

Figure 11: Valve Angle vs. Mass-flow on OX Valve

Another objective of the cold flow campaign was verifying controller performance under high pressure fluids, and especially with compressible fluids such as Liquid Oxygen. For this task the team first had to precisely measure and calibrate the discharge coefficient C_d for both injectors. The team used data from previous cold flows to get the closest estimate possible and tune the feedforward. Figure 12 shows a comparison between predictions based on the feedforward math and reality for an IPA cold-flow. Note that the delayed look of the real data is due to valve cavitation issues on throttle down and up. These issues were persistent across testing and will be discussed in more detail later.

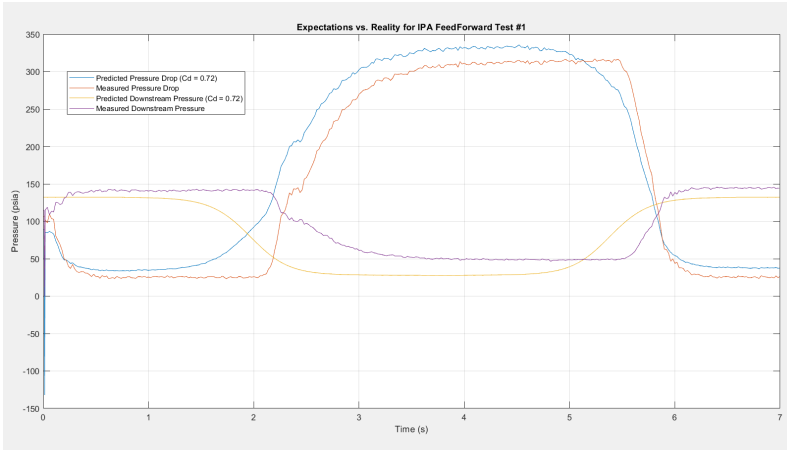

Figure 12: Cd Calibration on IPA Injector

The control capability verification consisted of multiple tests replicating the throttle curve on both valves (independently). In an effort to save propellant, these tests were only 7 seconds long instead of full-duration, except for two final verification runs which lasted for the whole 20 seconds. All of these tests but one were ran on closed loop control mode, as during the singular open loop test, the team experienced very strange behavior on the valves, where identical set pressures, temperatures, and valve angles produces wildly different mass-flows [Figure 13]

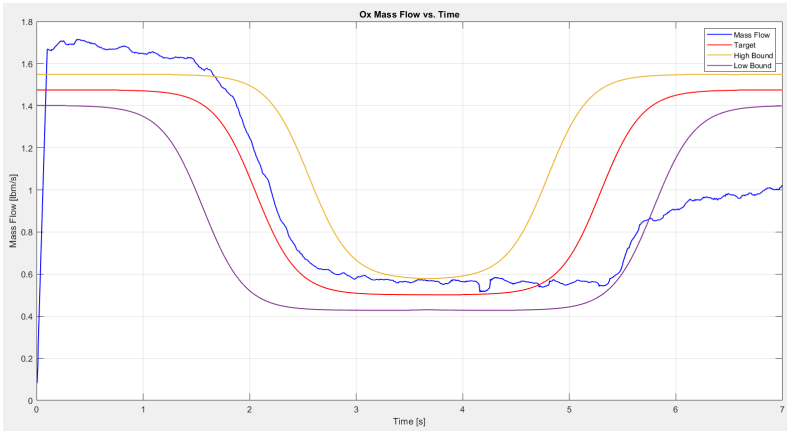

Figure 13: Cavitation of LOX Valve

The final control verification tests were satisfactory [Figure 14], proving good control over mass-flow under cryogenic and high pressure conditions, giving the team confidence to move forward with hot fire testing.
As clearly visible in Figure 14, the team struggled with increased noise on the IPA circuit mass-flow measurement. This has been attributed to mechanical vibrations in the area where the subcritical

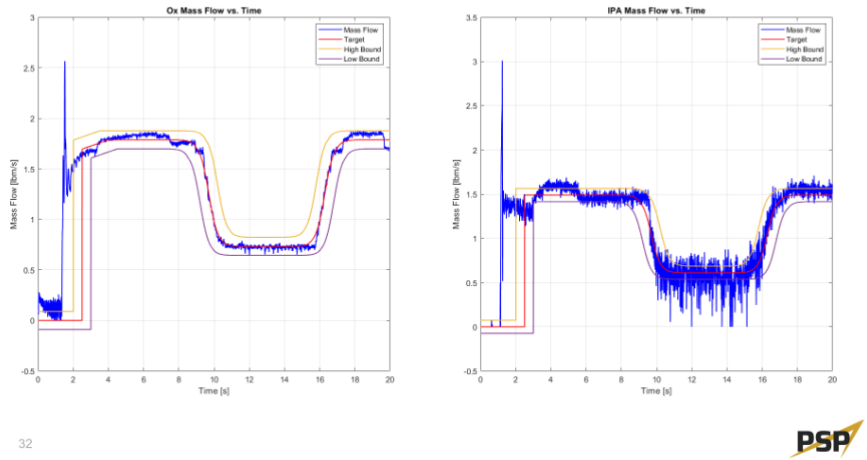

Figure 14: Control Verification

venturi was installed for IPA mass-flow measurement, on top of the fact that the differential pressure transducer for this mass-flow measurement had a 0-300 psi range, compared to just 0-50 psi for the LOX circuit. This led to increased noise too. Figure 15 shows a comparison of noise levels experienced between the measurements, by comparing the raw signal to a post filtered signal.

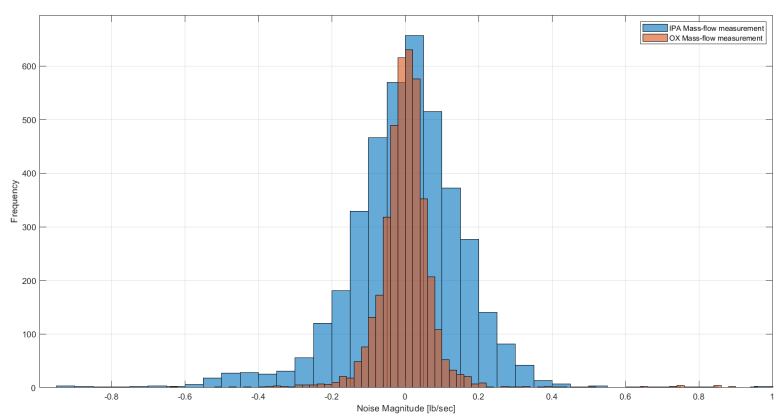

Figure 15: Noise Levels for both propellant circuits

Specifically, the standard deviation of the noise measured on the IPA circuit was 0.193 lb/sec against 0.098 lb/sec on the OX circuit. As most of the control action was provided by the feed-forward controller, and the rest was almost exclusively integral control, this noise had no significant effect on the controller performance.

7 Hotfire Campaign

The hot-fire campaign was the culmination of months upon months of work for every member of the team. After a successful completion of static, full duration hotfire tests at both 100% and 40% throttle using cavitating venturis for mass-flow control, it was time to complete the CPLC Milestone. To serve this purpose, the team attempted a series of tests to get ready for the attempt. Initially, short burp tests were conducted to verify the new ignition sequence using throttle valves, subsequently, the controller hand-off process was verified and tested, with an 8 second max thrust hold. It was here that the team started to notice some issues, Specifically, valve tracking performance had significantly deteriorated from previous tests, this caused problems with the integral wind-up. An example of poor tracking performance during a hotfire is shown in Figure 16.

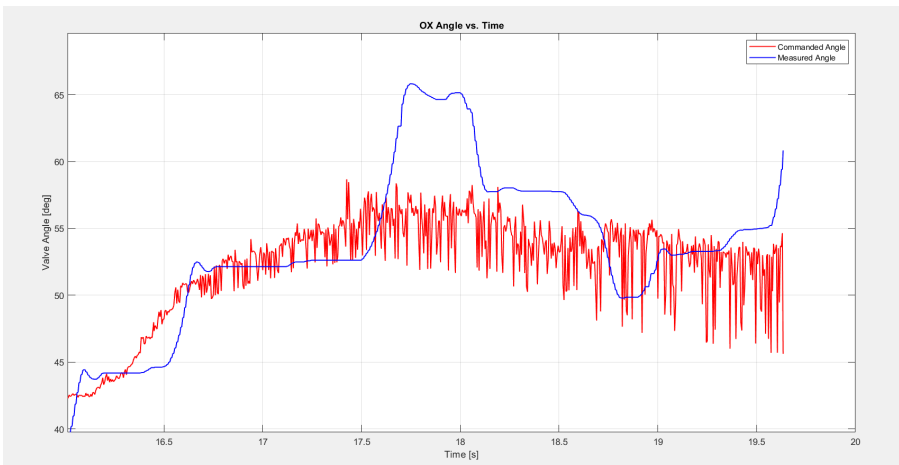

Figure 16: Noise Levels for both propellant circuits

Deviations as extreme as 10 degrees of position error were encountered. On top of this, both valves were found to be sticky, and required a variable minimum torque for movement to occur, which also caused problems with controller wind-up. Previously, the team had already implemented wind-up protection in the form of integral action clamping, nevertheless this method was mostly for safety and trim controller action limitation, as to avoid insane command values. It was challenging to make the clamping tighter, because due to the nonlinearity of the system, the trim controller needed as much control authority as possible, and clamping it more would reduce it more than possible. To deal with the wind-up concerns, the team implemented wind-up protection in the form of back-calculation. A standard integral gain in a PID controller accumulates as follows:

$$I = K_i \cdot \int_0^t e(\tau)\, d\tau$$

The back-calculation implemented for TADPOLE was of the form

$$I = K_i \cdot \frac{|\epsilon_{act}(t)|}{T} \cdot \int_0^t e(\tau)\, d\tau$$

In which $\epsilon_{act}(t)$ is the error between commanded actuator position and measured actuator position, and T is a selected tolerance range, usually set to match the actuator out of range abort scenario. (In TADPOLE, the test would be aborted if the angle position deviated by more than 8 degrees from commanded). This method of back-calculating is not standard, as back-calculation is commonly applied to saturated actuators, instead of stuck actuators. Nevertheless, this formulation causes a reduced accumulation proportional to how lagged behind the inner loop is. In essence, it makes the outer loop patient, and waits until the inner loop (in this case, the actuator control loop) reacts to begin accumulation again, and not overcompensate. This hot-fix proved extremely useful, and it is regarded as one of the most important lines of code added to the software stack during the campaign.

As far as the valve performance and stickiness, the team attempted to retune the valves under pressure load from active N_2 purges. Even with this, the valves were still behaving sub optimally, and due to the variable minimum torque for movement, the use of a feedforward on the torque controller was off the table. As such, the team decided to purposefully overtune the valves, as to cause controlled, low amplitude oscillations, essentially replicating the addition of a Dither signal to create constant valve movement and avoid sticking. Figure 17 shows this signal on our IPA valve during a hotfire test. Due to the high frequency nature of the oscillations, effects on mass-flow and flow patterns were negligible.

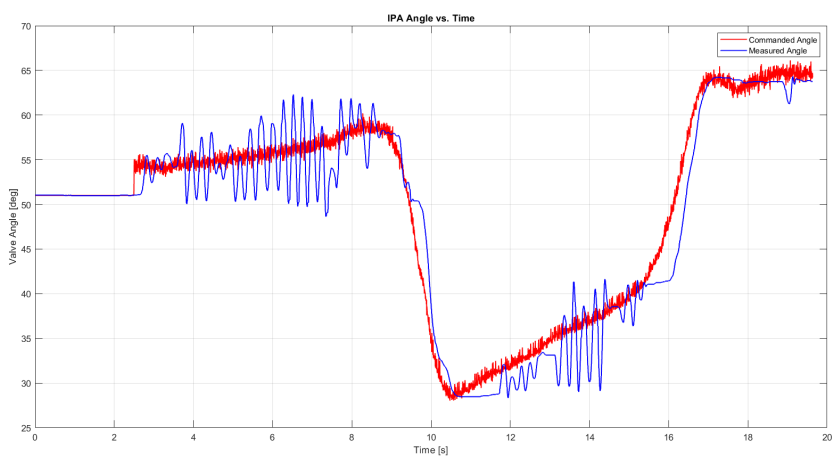

Figure 17: IPA Valve Tracking during Hotfire

This solution caused a noticeable improvement in valve tracking, and although not perfect, it proved enough for the milestone attempts. Overall, it took 3 attempts to complete the milestone. The first attempt was sub optimal. This attempt was previous to implementation of both the back-calculation and dither signal, so valve performance was abysmal and integral windup proved horrible. This, mixed in with the yet unsolved valve cavitation problems caused poor tracking of chamber pressure, and caused massive spikes on OF ratio, passing through stoichiometric mixture ratio three times. This behavior prompted a manual abort after 13 seconds of test. After

subsequent injector health checks by flowing nitrogen and checking that $C_d \cdot A$ were within 1% of nominal. The second attempt was significantly better, fully completing the curve with proper tracking. Nevertheless, due to the lack of a chamber pressure trim controller, and unexpected combustion behavior (c^* efficiency improving at lower thrust values), the thrust values were outside of the required bounds. The team created an interpolation for c^* values depending on commanded thrust to correct the feedforward, and tried again an hour later. This 3rd attempt was the first success on the CPLC milestone curve. Figure 18 and 19 show both the throttle curve for the attempt and mass flow tracking for both propellants.

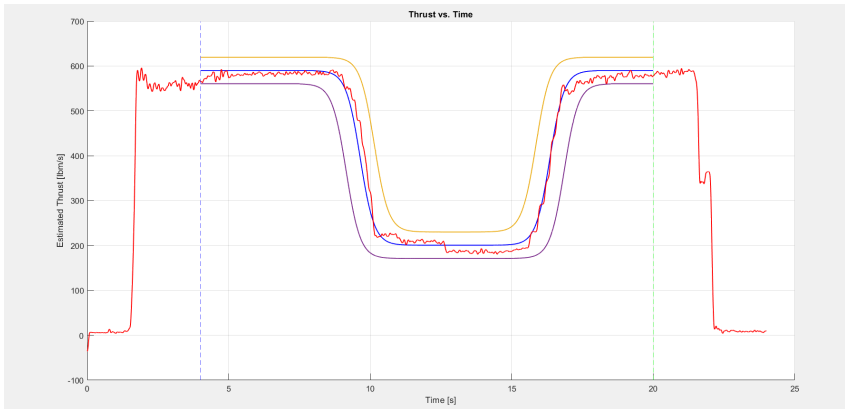

Figure 18: Final thrust curve submitted for the CPLC Milestone

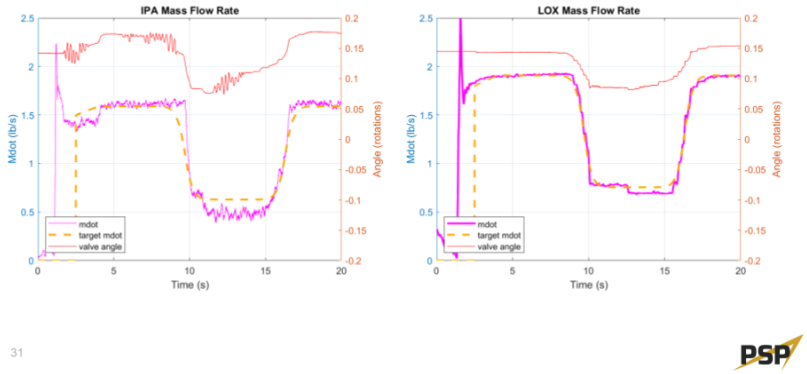

Figure 19: Mass-flow and valve tracking for both propellant circuits

An unresolved issue during the campaign was the previously mentioned valve cavitation on throttle up and throttle down. This issue was present on both valves, nevertheless, it wasn't symmetrical, This caused significant spikes in OF ratio on throttle up in almost every test. In future testing campaigns, the team is considering switching the trim controllers from mass-flow to OF ratio di-

rectly, as well as improving the hardware. Finally, the addition of another back calculation on the Chamber Pressure controller (by correcting measured mass flow vs target mass flow) would decouple the inner loop and allow for proper control without fear of oscillations.

Overall, the throttle campaign produced 9 hotfire attempts in just two days of testing, proving the ability to rapidly test, recycle, iterate, and produce results. This success demonstrates the dedication of the whole team, setting us up for success as we continue to work towards our full flight vehicle and final goal of producing a vertical take-off, vertical landing rocket, named TOAD.